

UNIT - III

Alternative building-materials for low-cost housing.

Introduction :-

→ Low-cost building is a critical global challenge

Importance of low-cost housing :-

- Affordability
- Improving local conditions
- Sustainability
- Economic development

Demand for low-cost housing :-

- cost savings
- over population
- migration from rural to urban.
- poverty
- increasing land costs

Alternative building-materials

→ Alternative building-materials are diverse range of construction materials other than traditional construction materials like concrete, wood, steel.

→ These construction materials are

- * natural
- * recycled &
- * engineered

→ Eco-friendly to the environment.

→ If we use these construction materials there can be lower construction costs.

→ some examples of Alternative building materials:-

such as bamboo, recycled plastic, straw bales, wood, Green concrete, Adobe, cork, straw, rammed earth.

→ These materials are

1) reusable.

2) recyclable

3) natural

4) durable.

5) low cost

Advantages of Alternative building materials.

1] Affordability → materials with locally sourced, less expensive and to produce low-cost housing.

2] sustainability →
→ reduce wastages
→ recyclable
→ reusable
→ does not impact on

3] Energy efficiency ÷ some of the ^{environment.} alternative building materials (like straw bales, rammed earth) offers good insulation properties.

4] local sourcing ÷

→ These materials can be available in local areas, and surroundings

→ it reduces the transportation costs.

5] community involvement ÷ Important aspect challenges and considerations.

→ ensuring structural integrity & durability.

→ Adequate training & skills for developers & builders.

→ community involvement

→ regulatory framework

→ Zoning hurdles.

substitute for scarce material

→ "substitute for scarce materials" refers to finding the alternative materials or resources to replace the materials which are in limited supply or scarcity ^{facing}.

→ This concept is mainly related to industries & technologies.

→ where, some times certain raw-materials are not available in surroundings.

→ Instead of that raw materials, we are choosing alternative materials.

* TO reduce transportation costs

* production costs

* material costs.

* low-quality
* less durability
* not locally available

→ finding a substitutes for scarce material is a complex process it involves many adv & disadvantages.

Advantages of substitute scarce materials:

Advantages :-

1] Reduce material costs.

2] Transportation costs (we use local sourcing materials)

3] production costs.

4] environmental benefits

5] Resource diversification [

6] sustainability of materials ^{reduces the dependency}
7] reducing dependency

Dis-advantages :-

- quality
- less performance
- market uncertainty
- compatibility issues [sometimes substitute may not meet the existing infrastructure]
- Technical challenges [identifying the substitutes are technically challenging task].
- Research & development costs :-

- * time consuming

- * developing new materials are complex.

- * skilled engineers.

Applications of substituting scarce material

1] Renewable energy :- used in the
wind turbines
solar panels
batteries.

2] Aerospace :- low-weight
high-strength materials.

3] electronics.

4] Automotive industries

5] medical devices.

6] construction

→ The main objective of this scarce is
reduce costs.

→ some of the materials → manganese

* vanadium. * carbon radiators → chromium

* Gallium. → Tungsten

* Nickel → Aluminium radiators

Ferro cement = framework of mesh or metal reinforcement

→ ferro cement is a composite material which can be used in a construction.

→ Ferro cement is & consists the thin-layer cement and extra framework of mesh or metal reinforcement.

→ ferro cements has unique properties

→ it is known for the

* durability

* strength

* versatility.

→ flexibility in shaping.

→ it has more strength compared to the normal cement.

Advantages

1] strength

2] durability

3] Versatility [formed into different form & sizes]

4] Light weight [easy to handle during work
Easy to transport]

5] fire resistance

6] cost effective

8] water proof

7] crack-resistance

9] earthquake resistance.

Dis-advantages

1] skill dependent

2] quality control

3] Performance.

4] labour requirement [layering]

5] maintenance

Applications

mainly used in architectural & industrial construction.

- 1] water tanks
- 2] Building construction
- 3] Bridge construction
- 4] Architectural structures
- 5] swimming pools.
- 6] Boats
- 7] shelves
- 8] low-cost housing.
- 9] Agriculture

Gypsum boards.



- Gypsum boards are also known as dry wall or plaster boards.
- It is type of material used in construction material.
- It consists of core gypsum (natural mineral) sandwiched b/w two layers of papers or fiberglass.
- These boards are used in various building applications because of:
 - ease of installation
 - versatility.
 - fire resistance.
- These are mostly used in interior designs in commercial residential industrial buildings etc.

Advantages :-

- 1] ease of installation [light-weight]
- 2] versatility [different shapes]
- 3] fire resistance
- 4] sound insulation
- 5] less cost
- 6] sustainability
- 7] smooth finish.

disadvantages :-

- 1] limited load-bearing capacity
 - 2] Less strength
 - 3] Easily breakable
 - 4] vulnerability in moisture
 - 5] fragile edges
- extra reinforcement tape joints

Applications :-

- ① Interior wall cladding
- ② ceiling finishes
- ③ sound proofing
- ④ Decorative purposes
- ⑤ Fire-rated wall
- ⑥ Commercial
- ⑦ Residential
- ⑧ Industrial

Timber substitution :-

It refers to the replacing the traditional timber or wood materials with alternative materials,
→ due to concern about deforestation,

* Sustainability

* Availability of wood materials.

→ This is an important aspect of reducing the environmental impact.

forests

deforestations

Advantages :

→ reduce the environmental impact

→ deforestation

→ Pollution

→ sustainability

→ Customization [timber substitution may get the new designers]

→ Lower maintenance is required.

→ stability [more stable compared to traditional wood]

→ Durability [

→ consistency.

disadvantages :

(1) cost (more cost)

(2) Transportation challenges

(3) Aesthetic [alternative materials may not give the superior design]

(4) Limited Availability.

(5) Increases material cost.

Applications

1] construction

2] flooring

3] Roofing

4] ceiling

5] Land scaping

6] Furniture.

7] marine & coastal constructions.

Industrial wastes :-

→ Industrial wastes refers to the Industrial byproducts } which are
Industrial solid wastes } materials
generated by the result of industrial processing
manufacturing
industrial activities

→ The companies which are encouraged to the recycling, reuse, adapting waste products can reduce the these industrial wastes.

→ Industrial waste can be

- 1) solid
- 2) semi-solid
- 3) liquid

→ Industrial waste can be hazardous & Non-hazardous.

Ex :- chemicals, oil, solvents, metals, vegetable waste from restaurants.

→ If the waste can dumped into seas or rivers it leads various damages to public health

→ Some industries produce bio-degradable wastes such as

- 1) paper
- 2) sugar
- 3) food
- 4) wood.

Advantages of industrial wastes

- 1] cost reduction
- 2] Recycling, Repurposing
- 3] produces energy
- 4] creates employment

- Improves efficiency
- conserves resources.
- Innovations.

Disadvantages :-

- 1) Environmental impact.
- 2) water pollution, soil pollution.
- 3) Air-pollution
- 4) public health issues.
- 5) Liability

Applications of Industrial waste

- 1] Recycling
- 2] Restaurants
- 3] Industries
- 4] manufacturing material.
- 5] Chemical manufacturing.
- 6] construction materials.

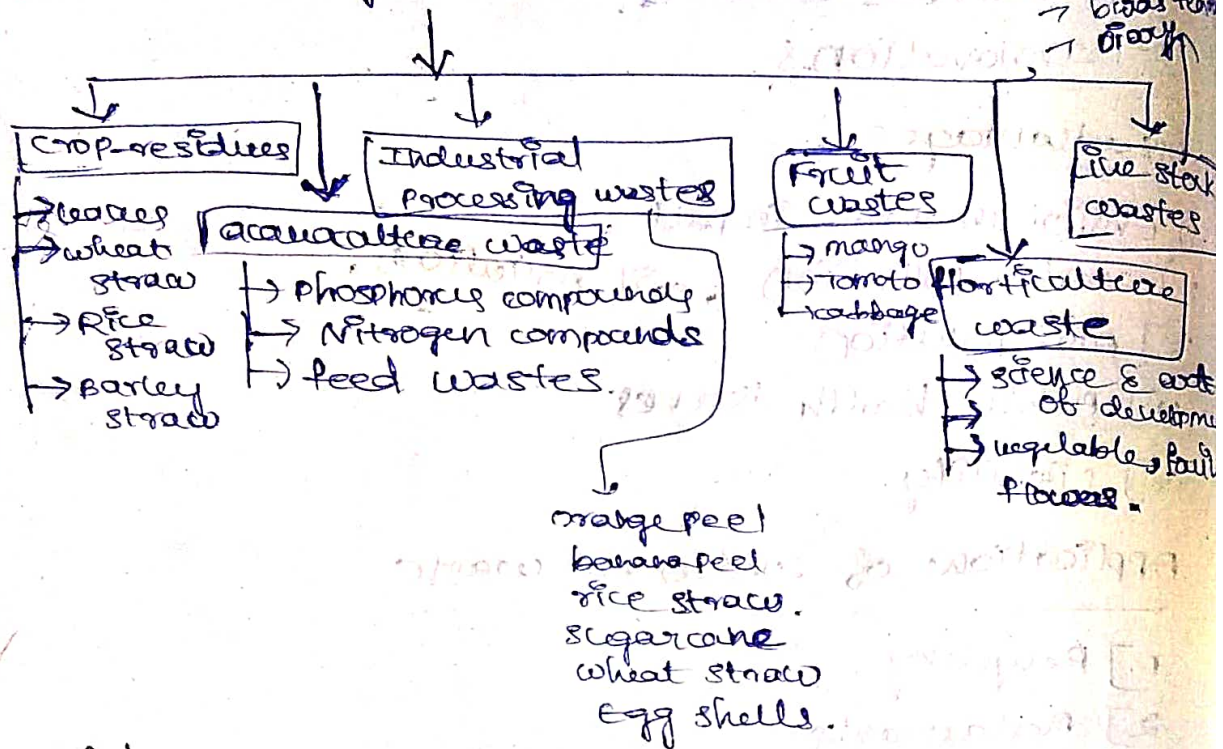
Agricultural waste :-

→ Agricultural waste refers to the byproducts or waste materials generated by the various Agricultural activities.

→ This can have both advantages & dis-advantages characteristics of agricultural waste

- 1] Physical properties
- 2] Bio-degradability
- 3]

Agriculture ¹²³⁴Waste



Advantages :-

- 1] composting :- improves the soil properties
 - 2] Bio-fuel :- plant waste can be used for producing Bio-fuel ; Bio-diesel .
 - 3] ~~Agriculture feed~~ :- Animal feed - agriculture waste can be used for animal feed .
 - 4] Bio-energy :- Agriculture waste can be used to create Bio-energy .
→ gobar gas .
 - 5] chemicals :- Agriculture waste → enzymes , Vitamins , antibiotics .
- Reducing green house gases
 - Reducing cost of waste disposal .
 - Reducing pollution
 - creating jobs .

disadvantages of agricultural wastes

negative effects on environment

- ① Air pollution [fertilizers, insecticides, heavy machinery]
- ② water pollution
- ③ soil pollution [decrease soil fertility]
- ④ greenhouse gas emissions [Agri waste can create the greenhouse gas emissions like CO₂, methane, nitrous oxide]
- ⑤ unpleasant odors
- ⑥ toxic liquids [infiltrate waters]

If that waste not disposed properly, plastic and packages can takes 100 years to decompose

Applications

- composting [fertilizer, animal feed]
- Bio-fuel production [plant waste]
- Reverse osmosis [remove dyes, color]
- manure management systems
- Bio-diesel production

Alternative building maintenance

- Alternative building maintenance refers to the another materials, sustainable & eco-friendly approaches used for repair buildings
- These materials can
 - * reduce environmental impact
 - * increase energy efficiency
 - * provide long-term sustainability

→ Alternative building materials like

* bamboo → local sourcing

* cork

* straw

* Adobe

* Green cement

→ Reduce, reuse, recycle

→ Reduce pollution.

Methods & practices

- ① using water efficiently.
- ② using sustainable materials.
- ③ using recyclable & renewable materials.
- ④ using ~~sustainable~~ environment-friendly materials.
- ⑤ Improving indoor air quality.
- ⑥ Reducing non-renewable energy consumption.

Advantages of Alternative building materials

- ① energy efficiency (has less energy & use passive design strategies & technologies to optimize performance)
- ② reduce water usage
- ③ sustainability (reduces environment impact)
- ④ Lower maintenance costs
- ⑤ Durability (longer life).
- ⑥ Reduced wastes.
- ⑦ less pollution.
- ⑧ ~~Health~~ Improves health (use of non-toxic materials)
- ⑨ increased productivity.

Disadvantages

- 1] Higher initial costs
- 2] skilled (or) experienced labourers
- 3] Limited Availability
- 4] unproven technologies (new techniques)
- 5] no-budgets control.

applications :-

- 1] Energy efficiency upgrades [insulation]
- 2] Air ceiling
- 3] weather ceiling
- 4] Solar-panel installation.
- 5] Non-toxic paints [improving indoor air quality]

low-cost infrastructure services :-

- low-cost infrastructure services refers to the construction of building can be done with "low-cost" without compromising the quality and performance.
- It is applicable for both public & private buildings.
- These services mainly used to reduce the ~~from~~ financial burden on individual business, Governments.
- These services are mostly used in utilities, commercial buildings, Presidential buildings, educational institutions, health care & more.
- The main objective :- no homeless people.
 - Homes can be available for All Spectrum of population.

key-points

- 1] Affordability [housing subsidies]
- 2] sustainability
- 3] Accessibility [low cost infra services it can access to all community]
- 4] efficiency [imp aspect for low-cost inf services]
- 5] public & private partnerships.

Examples of low-cost infrastructure services

- ① Affordable housing
- ② Public transportation
- ③ Healthcare [hospitals, medical trusts]
- ④ Education [scholarship programs to reduce the burden on acquiring knowledge]
- ⑤ Energy & utilization.

present status of the low-cost infra services

- based on region & country.
- A/c to my knowledge till 2023 updates regarding the low-cost infrastructure services

- 1] Housing → affordable housing
- 2] Public transportations
- 3] Healthcare
- 4] Education [online due to covid-19]
- 5] Transportation infrastructure [roads, bridges]
- 6] Telecommunication [payable]
- 7] Internet access [payable]
- 8] Public private partnerships [improves the infrastructure quality]
- 9] Energy & utilities [renewable energy resources have been reduced]
- 10] Water & sanitation [still these fields are developing].

Technological options

- Technological options play a crucial role in low-cost infrastructure services.

- 1) ~~to~~ improve the accessibility
- 2) increase the efficiency
- 3) Reduce operational costs

Key points

- Instead of using solar panel.
- 1] smart Grid [reduce operational costs]
 - 2] Renewable energy [reduce energy cost]
 - 3] Public transportation system [online booking]
 - 4] E-learning platform [online education]
 - 5] mobile banking
 - 6] online Payments
 - 7] waste management
 - 8] Digital health solutions
 - 9] Recycling Technologies.
 - 10] Agriculture Technologies
 - 11] Internet access.
 - 12] water purification.

~~Low-cost sanitation~~

Advantages :-

- 1] efficiency
- 2] sustainability
- 3] cost reductions [some of the technological options are Renewable]
- 4] quality improvements
- 5] Accessibility
- 6] Innovation [new Technologies]

Dis-advantages :-

- 1] cost Implementation
- 2] Digital Divide [for online class]
- 3] Data Privacy - security [more privacy & security for sensitive]
- 4] maintenance costs.
- 5] skills based [using these Technologies have much skills]
- 6]

Low-cost sanitation :-

- low-cost sanitation is critical aspect in low-cost infrastructure services.
- low-cost sanitation will ^{directly} ~~cause~~ impact the public health, environment and overall quality of life.
- Safe & affordable sanitation will reduce the spreading of diseases, reduce the environmental impact and ensures the well-being community.

Here some of strategies & technologies of

- 1] Improved pit toilets
- 2] ecosan Toilets
- 3] community & public toilet
- 4] urine diverting toilets
- 5] Biogas toilets
- 6] composting toilets
- 7] Low-flush toilets
- 8] Ventilated Improved pit toilets
- 9] mobile and containerized toilets.
- 10] monitoring & Data Analytics.
- 11] Decentralised wastewater treatment systems.

Advantages :-

- 1] Improves public health
- 2] environmental protection
- 3] spreading of diseases reducing
- 4] Resilience
- 5] community dignity (Access to all)
- 6] cost efficiency
- 7] Reduction in open defecation.

disadvantages of domestic wall

- maintenance cost
- Earthquake vulnerability
- quality control [high then only it can bear load]
- requires opening [material cost increase]
- Accessibility [wider door-way, ramps] Not visible
- Limited space
- Pre-casted walls [dis-advantages]

water supply

- It is an fundamental component of low-cost infrastructure services.
- And it is an important thing in human life mainly used for drinking and sanitation, and human daily life activities.
- Safe & clean water should be very important for public health and environmental protection.

considerations

- 1] source selection [wells, tanks, bore wells, ground water, surface water]
- 2] water treatment [filtration, purification, UV disinfection, chlorination]
- 3] distributed network [pipes, pumps]
- 4] decentralised systems
- 5] detection of leakage & repair
- 6] community participation [keep one person to maintain the water to their community]
- 7] Affordability
- 8] Ground water
- 9] Rain water harvesting.

Advantages of low-cost water supply

- | | |
|---------------------------|--------------------------|
| 1] Affordability | 5] community driven |
| 2] Improves public health | 6] helps the basic needs |
| 3] Improves sanitation | |
| 4] sustainability | |

dis-advantages ÷

- ① maintenance cost.
- ② operational cost.
- ③ water scarcity.
- ④ water leakage.
- ⑤ water quality.
- ⑥ initial investments.
- ⑦ Distribution of water.

Energy ÷

- energy is a crucial component in the low-cost infrastructure services.
- energy is the capacity to do something.
- it undergoes various aspects like
 - i] human activities -
 - ii] heating
 - iii] Transportation
 - iv] industrial processes.
- energy is available in various forms.
 - * light
 - * heat
 - * wind energy
 - * Thermal energy (heat energy)
 - * solar energy
 - * mechanical energy
 - * Gravitational energy.

consideration ÷

- 1) storage
- 2) supply
- 3) usage of energy
- 4] scarcity.
- 5] Biomass.
- 6] Bio-energy.
- 7] Decentralised systems:
- 8] community ownership

Advantages :-

- 1] Affordability
- 2] Environment sustainability
- 3] Energy efficiency
- 4] Economic Growth.
- 5] Rural development.

disadvantages :-

- initial installation
- Grid monitoring.
- Energy storage
- Energy scarcity.
- distribution of energy.
- cost efficiency.